

Retail FMCG Sales & Inventory Analysis

1. Project Overview

This project focuses on analysing **Retail FMCG (Fast-Moving Consumer Goods) sales, profitability, inventory, customer demographics, and sales channels** using Power BI.

The objective was to transform raw retail transaction data into an interactive business intelligence dashboard that helps businesses understand:

- Overall sales and profitability
- Category and brand performance
- Inventory availability and reorder requirements
- Customer demographics and loyalty
- Sales channel and payment behaviour
- City-wise sales performance
- Product pricing and profitability

The final Power BI dashboard consists of **four analytical pages**:

1. Overview
2. Product & Profitability
3. Inventory & Reorder
4. Customer & Channel Analysis

The dataset contains **100,000 retail transaction records** covering the year **2024**.

2. Business Problem

A retail FMCG business generates a large volume of daily sales transactions across different cities, store formats, product categories, brands, and sales channels.

Without proper analysis, it can be difficult for management to answer questions such as:

- How much revenue and profit the business is generating?
- Which product categories generate the most revenue?
- Which brands contribute the most profit?
- What is the overall profit margin?
- Which inventory categories have potential stock shortages?
- Which brands or categories require reordering?
- Which cities generate the most revenue?

- How does customer loyalty contribute to revenue?
- Which sales channels generate the most revenue?
- How are customer demographics distributed?
- What payment methods are being used?
- How do selling prices and costs relate to each other?

The purpose of this project was to build a centralized dashboard that provides these insights in an easy-to-understand format.

3. Project Objectives

The main objectives of the project were:

1. Analyse overall retail sales performance.
 2. Measure revenue, cost, profit, and profit margin.
 3. Identify high-performing categories and brands.
 4. Analyse selling price, cost price, and units sold.
 5. Monitor inventory levels against reorder levels.
 6. Identify transactions/brands requiring inventory attention.
 7. Analyse inventory by category and store format.
 8. Understand customer demographics.
 9. Compare loyalty and non-loyalty revenue.
 10. Analyse revenue across sales channels and payment methods.
 11. Compare city-level performance.
 12. Build an interactive Power BI dashboard for business decision-making.
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4. Dataset Information

Dataset: Indian FMCG Retail Sales Customer Inventory (2024)

Number of records: 100,000

Time period: 2024

Tool used: Microsoft Power BI

The dataset contains sales, pricing, inventory, customer demographic, and transaction-related information.

5. Dataset Columns and Their Meaning

COLUMN	MEANING
INVOICE_ID	Unique identifier associated with a sales invoice/transaction
INVOICE_DATE	Date on which the transaction occurred
CITY	City where the transaction/store activity occurred
STORE_FORMAT	Type or format of the retail store
CATEGORY	Product category
BRAND	Brand associated with the product
CHANNEL	Sales channel through which the transaction occurred
PAYMENT_MODE	Payment method used for the transaction
UNITS	Number of units sold
COST_PRICE	Cost of the product to the business
SELLING_PRICE	Price at which the product was sold
REVENUE	Revenue generated from the transaction
COST	Cost associated with the transaction
MARGIN	Profit/margin generated from the transaction
MARGIN_%	Percentage margin associated with the transaction
STOCK_ON_HAND	Available inventory/stock
REORDER_LEVEL	Inventory level at which replenishment should be considered
LEAD_TIME_DAYS	Number of days required for inventory replenishment
CUSTOMER_AGE	Age of the customer associated with the transaction
CUSTOMER_GENDER	Customer gender information
LOYALTY_FLAG	Indicates whether the customer is part of the loyalty program

6. Data Preparation and Cleaning

Before creating the dashboard, the data was examined and prepared for analysis.

Customer Age

The Customer_Age column contains missing values.

Instead of replacing missing numerical ages with arbitrary values, the original age field was retained as a numeric column.

A separate Age Group field was created for dashboard analysis:

- 18–25
- 26–35
- 36–45
- 46–55
- 56+
- Unknown

This allowed missing ages to be represented as **Unknown** without corrupting the numerical age data.

Customer Gender

The gender field contained:

- M
- F
- O
- Blank/missing values

For dashboard readability, a derived gender field was used to represent the categories as:

- Male
- Female
- Other
- Unknown

The Other category was retained rather than removing it.

Loyalty Flag

The original Loyalty_Flag field was represented as a user-friendly Loyalty Status field:

- Yes
- No

This makes the dashboard easier for business users to understand than displaying 1 and 0.

7. Data Analysis Measures

Several DAX measures were created to calculate important business KPIs.

Total Revenue

Total_Revenue = SUM(Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Revenue])

Measures the total revenue generated.

Total Profit

Total Profit = SUM(Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Margin])

Measures total profit/margin generated.

Profit Margin %

Margin_%_measure =

DIVIDE(
 [Total_Profit]
 [Total_Revenue],
 0
)

This calculates overall profit as a percentage of revenue.

A weighted calculation was used instead of simply summing or averaging the existing Margin_% values.

Total Units

Total_Units = SUM(Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Units])

Measures the total number of units sold.

Total Invoices

Total_Invoices =
DISTINCTCOUNT(Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Invoice_ID])

Counts distinct invoices.

Average Selling Price

Average_Selling_Price =
AVERAGE(Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Selling_Price])

Measures the average selling price.

Total Stock

Total_Stock = SUM(Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Stock_On_Hand])

Measures available stock.

Average Stock

Average_Stock =

AVERAGE(Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Stock_On_Hand])

Measures average stock level.

Average Lead Time

Average_Lead_Time =

AVERAGE(Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Lead_Time_Days])

Measures the average replenishment time.

Inventory Value

Inventory_Value =

SUMX(

Indian_FMCG_Retail_Sales_Customer_Inventory_(2024),

Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Stock_On_Hand] *

Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Cost_Price]

)

Estimates the value of available inventory based on stock and cost price.

8. Dashboard Page 1 — Overview

The first page provides a high-level overview of business performance.

KPIs

The dashboard displays:

- Total Revenue
- Total Profit
- Profit Margin %
- Units Sold
- Total Invoices

The final dashboard shows approximately:

- **Revenue:** ₹39.34M
- **Profit:** ₹7.87M
- **Profit Margin:** 20.02%

- **Units Sold:** 300K
- **Invoices:** 100K

These KPIs provide management with a quick overview of overall business performance.

Revenue & Profit Trend

A time-based chart was created to compare revenue and profit across the months of 2024.

This allows users to identify changes in sales and profitability over time.

Revenue by Category

Revenue was compared across product categories including:

- Fruits
- Grocery
- Snacks
- Beverages
- Home Care
- Vegetables
- Personal Care
- Dairy

Revenue by City

City-level revenue was analysed across cities including:

- Kolkata
- Delhi
- Hyderabad
- Chennai
- Pune
- Mumbai
- Ahmedabad
- Bengaluru

Revenue by Channel

Revenue was compared across:

- Online
- Offline
- Omnichannel

The dashboard shows the three channels contributing relatively similar shares of total revenue.

Top 10 Brands by Revenue

The dashboard also identifies the highest-revenue brands, including brands such as:

- ITC
- HUL
- Britannia
- Nestle
- Amul
- Tata
- PepsiCo
- Parle

Interactive Filters

The Overview page includes slicers for:

- Invoice Date
- City
- Store Format
- Category
- Channel

These allow users to interactively filter the dashboard.

9. Dashboard Page 2 — Product & Profitability

The second page focuses on product performance and profitability.

KPIs

The page contains:

- Total Revenue
- Total Profit
- Profit Margin %
- Total Cost
- Average Selling Price

The dashboard shows:

- Total Revenue: ₹39.34M

- Total Profit: ₹7.87M
- Total Cost: ₹31.46M
- Profit Margin: 20.02%
- Average Selling Price: approximately ₹131.11

Revenue vs Profit by Category

A combination chart compares revenue and profit across product categories.

This helps identify categories that contribute strongly to both sales and profitability.

Top 10 Brands by Profit

A Top 10 analysis was created using total profit.

This identifies brands contributing the highest total margin.

Selling Price vs Units

A scatter chart compares:

- Selling Price
- Units Sold

with product category used for segmentation.

This helps explore the relationship between pricing and sales volume.

Cost vs Selling Price

Another scatter chart compares:

- Cost Price
- Selling Price

This helps analyse the relationship between product cost and selling price.

Profit Margin % by Category

Profit margin was calculated using:

Total Profit ÷ Total Revenue

rather than summing the existing Margin_% column.

This provides a more meaningful category-level profitability comparison.

Category Performance Table

The table contains:

- Category
- Revenue
- Cost

- Profit/Margin
- Profit Margin %
- Units

This provides a detailed category-level performance summary.

10. Dashboard Page 3 — Inventory & Reorder

The third page focuses on inventory management.

KPIs

The page includes:

- Total Stock
- Average Stock
- Average Lead Time
- Inventory Value

The dashboard shows approximately:

- **Average Stock:** 274.16
- **Average Lead Time:** 8.52 days
- **Inventory Value:** ₹2.87B

Stock vs Reorder Level

A scatter chart compares:

- Stock on Hand
- Reorder Level

This helps identify inventory positions relative to their defined reorder thresholds.

Available Stock by Category

Stock levels were analysed across all product categories.

This helps identify categories holding larger or smaller amounts of inventory.

Reorder Required Table

A dedicated table was created containing:

- Brand
- Category
- Stock on Hand
- Reorder Level

- Lead Time
- Stock Status

A calculated field was used to classify stock:

Stock Status =

```
IF(
  Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Stock_On_Hand] <=
  Indian_FMCG_Retail_Sales_Customer_Inventory_(2024)[Reorder_Level],
  "Reorder Required",
  "Sufficient Stock"
)
```

The dashboard can then filter the table to display records requiring attention.

Lead Time by Category

Average lead time is analysed across product categories.

This helps understand which categories may require longer replenishment periods.

Inventory by Store Format

Inventory levels were compared across store formats such as:

- Hyper
- Express
- Super

11. Dashboard Page 4 — Customer & Channel Analysis

The fourth page focuses on customer demographics, loyalty, sales channels, cities, and payment methods.

Revenue by Gender

Revenue was analysed across:

- Female
- Male
- Other
- Unknow

The dashboard shows approximately:

- Female: ₹17.67M

- Male: ₹17.66M
- Other: ₹2.03M
- Unknown: ₹1.9M

Loyalty vs Non-Loyalty Revenue

Revenue was compared between loyalty and non-loyalty customers.

The dashboard shows approximately:

- Non-Loyal: ₹27.58M
- Loyal: ₹11.75M

The loyalty revenue represents approximately **29.88% of total revenue**.

Revenue by Age Group

Customers were grouped into:

- 18–25
- 26–35
- 36–45
- 46–55
- 56+
- Unknown

This allows the business to compare revenue contribution across age groups while keeping missing ages separate.

Revenue by Channel

Revenue was compared across:

- Omnichannel
- Online
- Offline

Revenue by City

Revenue was analysed across the available cities.

This allows management to compare geographic sales performance.

Average Customer Age

The dashboard calculates an average customer age of approximately:

41.08 years

Payment Mode

Revenue was analyzed across payment methods including:

- UPI
- Wallet
- Card
- Cash

The dashboard shows relatively similar revenue contributions across the payment methods.

12. Key Business Findings

Based on the completed dashboard:

Overall Performance

The dataset represents approximately **₹39.34 million in revenue** and **₹7.87 million in profit**, producing an overall profit margin of approximately **20.02%**.

Category Performance

Revenue is distributed across eight major FMCG categories, with no single category overwhelmingly dominating the dataset. Category-level analysis allows the business to compare both revenue and profitability.

Brand Performance

The Top 10 brand analysis identifies brands contributing strongly to revenue and total profit.

Inventory

The dashboard identifies inventory positions relative to reorder levels and provides a dedicated reorder-required view.

Customer Loyalty

Approximately **29.88% of total revenue** comes from loyalty customers based on the dashboard calculations.

Sales Channels

Online, Offline, and Omnichannel sales each contribute roughly one-third of total revenue in the dataset.

Customer Demographics

The dashboard provides demographic analysis using gender and age groups while keeping missing demographic information separate rather than artificially assigning values.

Payment Methods

UPI, Wallet, Card, and Cash all contribute meaningful portions of transaction revenue.

13. Tools & Technologies Used

Power BI

- Dashboard development
- Data visualization
- Interactive filtering
- KPI cards
- Charts and tables
- DAX measures
- Business analysis

Power Query

- Data preparation
- Data cleaning
- Handling missing values
- Data transformation

DAX

Used to create calculated measures and business metrics such as:

- Revenue
- Cost
- Profit
- Profit Margin %
- Total Units
- Invoice Count
- Average Stock
- Average Lead Time
- Inventory Value
- Loyalty Revenue

14. Dashboard Visuals Used

The project uses multiple Power BI visual types:

- KPI Cards
- Line/Column charts

- Bar charts
- Donut charts
- Scatter charts
- Tables
- Slicers

These visuals were selected according to the business question being analyzed.

15. Business Value of the Dashboard

The dashboard can help a retail business:

- Monitor overall sales performance
- Track profitability
- Compare product categories
- Identify high-performing brands
- Monitor inventory levels
- Identify stock requiring reorder
- Understand customer demographics
- Analyse loyalty-program revenue
- Compare sales channels
- Analyse payment methods
- Compare city performance
- Support data-driven business decisions

Instead of manually analysing large transaction datasets, management can use the dashboard to interactively explore performance using filters and visualizations.

16. Project Outcome

The raw FMCG transaction data was transformed into a **four-page interactive Power BI business intelligence dashboard** covering:

Sales → Profitability → Inventory → Customers → Channels

The project demonstrates an end-to-end analytics workflow involving:

Raw Data → Data Cleaning → Data Transformation → DAX Measures → Data Analysis → Visualization → Business Insights

This project demonstrates practical skills in **Power BI, Power Query, DAX, data cleaning, data visualization, KPI development, and business-oriented data analysis.**

17. Conclusion

The Retail FMCG Sales & Inventory Analysis project provides a consolidated view of retail business performance for 2024.

The dashboard combines financial, product, inventory, customer, geographic, channel, and payment information into a single interactive reporting solution.

The analysis demonstrates how Power BI can transform a large transactional dataset into a business-friendly dashboard that helps stakeholders monitor performance, identify areas requiring attention, and explore sales and inventory patterns interactively.